

IRE 317

Control System Engineering

Sadia Enam

Assistant Professor, IRE

University of Frontier Technology, Bangladesh

Error Analysis

Pole and Zero of Control System:

A **transfer function** $G(s)$ of a linear time-invariant (LTI) system is generally written as:

$$G(s) = \frac{N(s)}{D(s)} = \frac{(s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)}$$

Where:

$N(s)$:numerator polynomial

$D(s)$:denominator polynomial

$z_1, z_2, \dots, z_m$:**zeros**

$p_1, p_2, \dots, p_n$:**poles**

Error Analysis

Pole and Zero of Control System:

Zeros:

The values of s that make the numerator zero i.e $N(s)=0$

They are the frequencies where the output becomes zero (system response is nullified).

Poles:

The values of s that make the denominator zero i.e $D(s)=0$

They determine the stability and dynamic behavior of the system

For the transfer function

$$G(s) = \frac{s + 2}{(s + 1)(s + 3)}$$

Zero: $s = -2$

Poles: $s = -1, -3$

The zero occurs where the numerator is zero, while the poles occur where the denominator is zero. The poles determine whether the system is stable, whereas the zero affects how the system responds to an input.

Error Analysis

You have a machine (system).

You give it an input, and it gives you an output.

Now, for some inputs, the machine works normally but for some special inputs, the machine gives no output at all (output = 0).

The point where the output becomes zero is called a Zero of the system.

Let's take a system:

$$G(s) = \frac{s - 2}{s + 3}$$

Here,

Numerator = $s - 2$

Denominator = $s + 3$

Now, if we put $s = 2$:

$$G(s) = \frac{2 - 2}{2 + 3} = \frac{0}{5} = 0$$

That means the system's output becomes 0

Even though we give an input, the system produces nothing.

So,

$$s = 2$$

is called a Zero, because the system's response becomes zero at that point.

Error Analysis

Pole:

Poles decide if the system is stable or unstable.

They control how the system reacts over time — if it settles quickly, slowly, or oscillates.

If poles are in the “bad” place (right side), the system can get out of control.

Zero:

Zeros change the shape of the system’s output.

They can make the output zero at some points or reduce it.

Zeros help control how smooth or bumpy the response is.

Error Analysis

Type 0, Type 1, and Type 2 control systems, which are based on how many poles are at the origin ($s = 0$) in the open-loop transfer function

A general open-loop transfer function is:

$$G(s)H(s) = K \frac{N(s)}{D(s)}$$

Now we check how many poles are at the origin ($s = 0$).

That number = System Type.

System Type	Number of poles at origin ($s = 0$)
Type 0	0 poles at origin
Type 1	1 pole at origin
Type 2	2 poles at origin

Error Analysis

Classification of Control System:

Consider the open loop transfer function

$$G(s)H(s) = \frac{K(1+sT_1)(1+sT_2)\dots}{s^m(1+sT_a)(1+sT_b)+\dots} \quad \dots(3.1)$$

In equation (3.1), the poles are at $s = -\frac{1}{T_a}$, $s = -\frac{1}{T_b}$... and zeros are at $s = -1/T_1$,

$s = -1/T_2$... The equation having a term s^m in denominator, ' m ' is the number of poles at the origin.

A system having no pole at origin of the ' s ' plane, is said to be type '0' (zero) system i.e., $m = 0$

If $m = 1$ i.e., ' s ', it means the system has a pole at origin of the s -plane and is said to be type '1' (one) system.

A system is called type '2' system if $m = 2$. and so on.

Error Analysis

Steady state error: It is the difference between the input and output of the system during steady state.

For accuracy steady state error should be minimum.

We know that

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}$$
$$\therefore E(s) = \frac{R(s)}{1 + G(s)H(s)}$$

The steady state error of the system is obtained by final value theorem

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s.E(s)$$

Error Analysis

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1 + G(s)H(s)}$$

For unity feedback $H(s) = 1$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1 + G(s)}$$

Thus, the steady state error depends on the input and open loop transfer function.

Error Analysis

In control systems, when you apply inputs like step, ramp, or parabolic (acceleration) inputs, the system may not perfectly follow them. It may have some steady-state error (a small difference between desired output and actual output after a long time).

To measure how well a system tracks these inputs, we use error constants:

Position Error Constant K_p Measures system's ability to follow a step input (constant input).

Velocity Error Constant K_v Measures system's ability to follow a ramp input (input increasing linearly with time, like t).

Acceleration Error Constant K_a Measures system's ability to follow a parabolic input (input increasing quadratically with time, like t^2).

Error Analysis

STATIC ERROR COEFFICIENTS

STATIC POSITION ERROR CONSTANT K_p : For unit step input $R(s)=1/s$

Steady state error

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \cdot \frac{1}{1 + G(s)H(s)}$$
$$e_{ss} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)H(s)} = \frac{1}{1 + K_p}$$
$$K_p = \lim_{s \rightarrow 0} G(s)H(s)$$

Where K_p is the Static position error constant

Error Analysis

STATIC VELOCITY ERROR CONSTANT (K_v):

$$e_{ss} = \lim_{s \rightarrow 0} s.R(s) \cdot \frac{1}{1 + G(s)H(s)}$$

Steady state error with a unit ramp input is given by

$$R(s) = 1/s^2$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + G(s)H(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)H(s)}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{1}{sG(s)H(s)} = \frac{1}{K_v}$$

Where $K_v = \lim_{s \rightarrow 0} sG(s)H(s)$

Static velocity error coefficient

Error Analysis

STATIC ACCELERATION ERROR CONSTANT (K_a):

The steady state error of the system with unit parabolic input is given by

$$R(s) = \frac{1}{s^3}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s^3} \cdot \frac{1}{1 + G(s)H(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)H(s)}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{1}{s^2 G(s)H(s)} = \frac{1}{K_a}$$

where, $K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s)$ Static acceleration constant.

Error Analysis

STEADY STATE ERROR FOR DIFFERENT TYPE OF SYSTEMS

TYPE ZERO SYSTEM WITH UNIT STEP INPUT:

Consider open loop transfer function

$$G(s)H(s) = \frac{K(1+sT_1)(1+sT_2).....}{s^m(1+sT_a)(1+sT_b).....} \text{ --- (1)}$$

$$R(s) = \frac{1}{s}$$

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = K$$

$$\therefore e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+K}$$

$$\therefore e_{ss} = \frac{1}{1+K}$$

Hence , for type zero system the static position error constant K_p is finite.

Error Analysis

TYPE ZERO SYSTEM WITH UNIT RAMP INPUT:

$$K_v = \lim_{s \rightarrow 0} s G(s) H(s) = \lim_{s \rightarrow 0} s \cdot \frac{K(1+sT_1)(1+sT_2)\dots}{(1+sT_a)(1+sT_b)\dots} = 0$$

$$e_{ss} = \frac{1}{K_v} = \infty$$

$$\therefore e_{ss} = \infty$$

TYPE ZERO SYSTEM WITH UNIT PARABOLIC INPUT:

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) H(s) = \lim_{s \rightarrow 0} s^2 \cdot \frac{K(1+sT_1)(1+sT_2)\dots}{(1+sT_a)(1+sT_b)\dots} = 0$$

$$e_{ss} = \frac{1}{K_a}$$

$$\therefore e_{ss} = \infty$$

For type 0 system steady state error is infinite for ramp and parabolic inputs. Hence, the ramp and parabolic inputs are not acceptable.

Error Analysis

TYPE 'ONE' SYSTEM WITH UNIT STEP INPUT:

$$K_p = \lim_{s \rightarrow 0} G(s)H(s)$$

Put the value of $G(s)H(s)$ from eqn.1

$$K_p = \infty$$

$$e_{ss} = \frac{1}{1 + K_p} = 0$$

$$\therefore e_{ss} = 0$$

Consider open loop transfer function

$$G(s)H(s) = \frac{K(1+sT_1)(1+sT_2)\dots\dots\dots}{s^m(1+sT_a)(1+sT_b)\dots\dots\dots} \text{---(1)}$$

TYPE 'ONE' SYSTEM WITH UNIT RAMP INPUT:

$$K_v = \lim_{s \rightarrow 0} s.G(s)H(s)$$

Put the value of $G(s)H(s)$ from eqn.1

$$K_v = K$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{K}$$

$$\therefore e_{ss} = \frac{1}{K}$$

Error Analysis

TYPE 'ONE' SYSTEM WITH UNIT PARABOLIC INPUT:

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s)$$

Put the value of $G(s)H(s)$ from eqn.1

$$K_a = 0$$

$$e_{ss} = \frac{1}{K_a} = \infty$$

$$\therefore e_{ss} = \infty$$

Hence, it is clear that for type '1' system step input and ramp inputs are acceptable but parabolic is not.

Error Analysis

3 (a) Type '2' system with unit step input

$$G(s)H(s) = \frac{K(1+sT_1)(1+sT_2)\dots}{s^2(1+sT_a)(1+sT_b)\dots}$$

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{K(1+sT_1)(1+sT_2)\dots}{s^2(1+sT_a)(1+sT_b)\dots} = \infty$$

$$e_{ss} = \frac{1}{1+K_p} = 0 \quad \boxed{e_{ss} = 0}$$

(b) Type '2' system with unit ramp input

$$K_v = \lim_{s \rightarrow 0} sG(s)H(s)$$

$$= \lim_{s \rightarrow 0} s \cdot \frac{K(1+sT_1)(1+sT_2)\dots}{s^2(1+sT_a)(1+sT_b)\dots}$$

$$= \infty$$

$$e_{ss} = \frac{1}{K_v} = 0 \quad \boxed{e_{ss} = 0}$$

Type '2' system with unit parabolic input

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s)$$

$$= \lim_{s \rightarrow 0} s^2 \cdot \frac{K(1+sT_1)(1+sT_2)\dots}{s^2(1+sT_a)(1+sT_b)\dots} = K$$

$$e_{ss} = \frac{1}{K_a} = \frac{1}{K} \quad \boxed{e_{ss} = \frac{1}{K}}$$

Hence, for type '2' system all three inputs

Error Analysis

INPUT SIGNALS	TYPE '0' SYSTEM	TYPE '1' SYSTEM	TYPE '2' SYSTEM
UNIT STEP INPUT	$\frac{1}{1+K}$	0	0
UNIT RAMP INPUT	∞	$\frac{1}{K}$	0
UNIT PARABOLIC INPUT	∞	∞	$\frac{1}{K}$

Error Analysis

The open loop TF of unity feedback system is given by $G(s) = \frac{50}{(1+0.1s)(s+10)}$. Determine static error coefficients K_p , K_v , K_a

$$K_p = \lim_{s \rightarrow 0} G(s)$$

$$K_v = \lim_{s \rightarrow 0} sG(s)$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)$$

$$K_p = \lim_{s \rightarrow 0} \frac{50}{(1 + 0.1s)(s + 10)}$$

Substitute $s = 0$:

$$K_p = \frac{50}{(1)(10)} = 5$$

$$K_v = \lim_{s \rightarrow 0} \frac{50s}{(1 + 0.1s)(s + 10)}$$

Substitute $s = 0$:

$$K_v = \frac{50(0)}{(1)(10)} = 0$$

$$K_a = \lim_{s \rightarrow 0} \frac{50s^2}{(1 + 0.1s)(s + 10)}$$

Substitute $s = 0$:

$$K_a = \frac{50(0)^2}{(1)(10)} = 0$$

Error Analysis

The forward path TF of unity feedback system is given by $G(s)$
 $= \frac{5(s^2+2s+10)}{s^2(s+5)(s^2+3s+10)}$. Determine static error coefficients K_p , K_v , K_a

Error Analysis

The block diagram of an electronic pacemaker is given in following figure. Determine the steady state error for unit ramp input when $k=400$. Also, determine the value of k for which the steady state error to a unit ramp will be 0.02.

Solution: Given that $K = 400$

$$R(s) = \frac{1}{s^2}$$

$$H(s) = 1$$

$$G(s)H(s) = \frac{K}{s(s+20)}$$

Steady state error is given by $e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1+G(s)H(s)}$

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{K}{s(s+20)}} = \lim_{s \rightarrow 0} \frac{s+20}{s(s+20) + 400} = 0.05 \quad \text{Ans.}$$

Now $e_{ss} = 0.02$ (given)

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{1}{s^2} \cdot \frac{1}{1 + \frac{K}{s(s+20)}}$$

$$0.02 = \lim_{s \rightarrow 0} \frac{s+20}{s(s+20) + K}$$

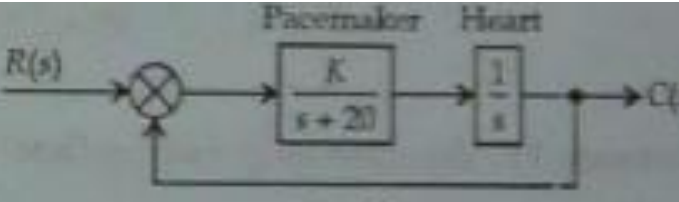
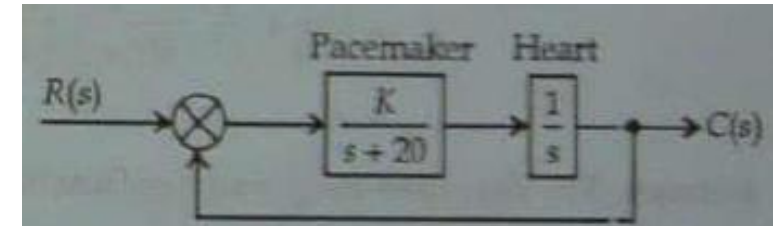
$$0.02 = \frac{20}{K} \quad \therefore \quad \boxed{K = 1000} \quad \text{Ans.}$$


Fig. 3.2



Error Analysis

For a unity feedback control system, the forward path TF is given by $G(S) = \frac{20}{s(s+2)(s^2+2s+20)}$. Determine the steady state error of the system. When the inputs are (i) 5, (ii) $5t$ (iii) $\frac{3t^2}{2}$

$$K_p = \lim_{s \rightarrow 0} G(s)$$

Since $G(s)$ contains $1/s$,

$$\begin{aligned} K_p &= \infty \\ K_v &= \lim_{s \rightarrow 0} \frac{sG(s)}{20} \\ &= \lim_{s \rightarrow 0} \frac{20}{(s+2)(s^2+2s+20)} \end{aligned}$$

Substituting $s = 0$:

$$K_v = \frac{20}{2 \times 20} = \frac{20}{40} = 0.5$$

$$\begin{aligned} K_a &= \lim_{s \rightarrow 0} \frac{s^2 G(s)}{20s} \\ &= \lim_{s \rightarrow 0} \frac{20s}{(s+2)(s^2+2s+20)} = 0 \end{aligned}$$

Error Analysis

For a unity feedback control system, the forward path TF is given by $G(S) = \frac{20}{s(s+2)(s^2+2s+20)}$. Determine the steady state error of the system. When the inputs are (i) 5, (ii) 5t (iii) $\frac{3t^2}{2}$

(i) Input: $r(t) = 5$ (Step Input)

$$R(s)=5/s$$

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} s \frac{R(s)}{1+G(s)H(s)} = \lim_{s \rightarrow 0} s * \frac{5}{s} \frac{1}{1+\frac{20}{s(s+2)(s^2+2s+20)}} = 0$$

(ii) Input: $r(t) = 5t$ (Ramp Input)

$$R(s)=5/s^2$$

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} s \frac{R(s)}{1+G(s)H(s)} = \lim_{s \rightarrow 0} s * \frac{5}{s^2} \frac{1}{1+\frac{20}{s(s+2)(s^2+2s+20)}} = \lim_{s \rightarrow 0} s * \frac{5}{s^2} \frac{s(s+2)(s^2+2s+20)}{s(s+2)(s^2+2s+20)+20} = 10$$

(iii) Input: $r(t) = \frac{3t^2}{2}$ 5t(parabolic Input)

$$R(s)=3/s^3$$

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} s \frac{R(s)}{1+G(s)H(s)} = \lim_{s \rightarrow 0} s * \frac{3}{s^3} \frac{1}{1+\frac{20}{s(s+2)(s^2+2s+20)}} = \infty$$

Error Analysis

The open loop TF is given by $G(s) = \frac{108}{s^2(s+4)(s^2+3s+12)}$. Find the static error coefficients and steady state error of the system when subjected to an input given by $r(t) = 2 + 5t + 2t^2$